

7

Physical properties

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7.1 Overview

7.1.1 Crystal anisotropy

We are used to thinking about the physical properties of materials in terms of scalar quantities, such as volume and energy. These will vary as the external conditions are changed. Simple examples are the coefficient of volume expansion,

$$\beta = \frac{1}{V} \frac{\partial V}{\partial T} \quad (7.1)$$

or the compressibility,

$$K = \frac{1}{V} \frac{\partial V}{\partial P} \quad (7.2)$$

We also understand that some properties can have a vector form. For example, the magnetization of a ferromagnetic material is a vector quantity. Moreover, some external stimuli have the nature as vectors and give a response that can be described as a vector. An example is an electric field, $\mathbf{E}$, applied to a dielectric material giving an induced polarization field, $\mathbf{P}$. For an **isotropic** material (quite what we mean by ‘isotropic’ can remain ill-defined for the moment – it is sufficient to take the word at face value as implying no dependence of properties on direction) the direction of the response vector will be the same as that of the stimulus. Thus we might define a susceptibility χ as a simple scalar:

$$\frac{1}{\epsilon_0} \mathbf{P} = \chi \mathbf{E} \quad (7.3)$$

For many technological applications it is necessary to generalize this trivial approach to more complicated properties and to consider the effects of **anisotropy**, namely the dependence of properties on direction. Following our example, in an anisotropic material the application of a field $\mathbf{E}$ may result in a polarization $\mathbf{P}$ that is not parallel to $\mathbf{E}$. Thus we might write the components of the vector $\mathbf{P} = (P_1, P_2, P_3)$ piecewise in terms of the components of $\mathbf{E} = (E_1, E_2, E_3)$:

$$\frac{1}{\epsilon_0} \mathbf{P} = \chi \cdot \mathbf{E} \Rightarrow \frac{1}{\epsilon_0} P_i = \sum_{j} \chi_{ij} E_j \quad (7.4)$$

The susceptibility χ now has the form of a 3×3 matrix. Clearly now the response $\mathbf{P}$ depends on the direction of $\mathbf{E}$ in both its direction and magnitude, and the matrix of numbers that form the susceptibility determine the form of this dependence on direction.

So far the extension from the isotropic case seems relatively easy. Too easy in fact, because if we are not careful we will quickly end up in the wrong direction. An example of thermal expansion will demonstrate where we are heading. Consider a hexagonal material that expands along the $[001]$ direction on heating but shrinks along the $\langle 100 \rangle$ directions – this is not as unlikely as it might appear. You can imagine that it should be possible to cut the crystal into the shape of a plate with an orientation relative to the crystal axes such that there is an exact balance between the positive and negative contributions of the thermal expansion to give zero thermal expansion in the direction normal to the plate. This situation is illustrated in Fig. 7.1. The crystal axes, which define the main symmetry directions, are shown as the main axes, and the plate with zero thermal expansion normal to its face is shown in its appropriate orientation. To calculate the orientation of the cut faces of the plate relative to the crystal axes is not quite as trivial as you might first think. The proper description of thermal expansion is in terms of the concept of **strain**, which describes the change in both the size and shape of an object. To calculate how to cut a plate that does not expand on heating, we need to rotate the components of the thermal expansion, which means rotating the full mathematical form of the strain. This operation is not as simple as the rotation of a vector, although there are some common elements. In fact this type of problem is quite common, and the purpose of this chapter is to introduce the mathematical tools that have been developed to handle it. We will see that these tools are fairly powerful, and can be extended to determine the effects of symmetry on the physical properties of a crystal.

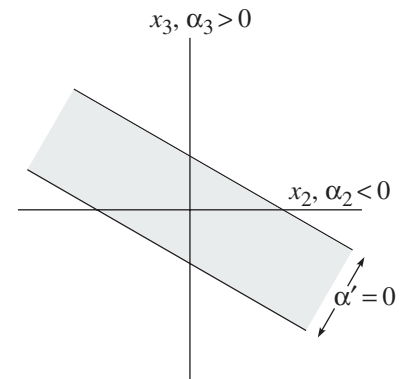


Fig. 7.1 Orientation of crystal plate with negative thermal expansion in one direction chosen so that the expansion of the plate is exactly zero.

7.1.2 An introduction to tensors

The mathematical form of the anisotropic properties of crystals is described using **tensors**. Vectors, such as those used to describe the dielectric polarization, and matrices, such as those used to describe the susceptibility, are special forms of tensors, and we will quickly meet higher-order versions. Tensors are rather more than mere collections of numbers. Integral to the definition of a tensor is the ability to change the definition of the coordinate system, the operation that we encountered in the example of the crystal plate with zero net thermal expansion. Since the physical properties of a crystal do not depend on our choice of axes, we expect the transformation tools for the rotation of the axes to leave the physics intact. A significant part of this chapter is to understand how to apply such transformations.

The vector tensors such as polarization are called **first-rank tensors**, and the matrix tensors such as susceptibility are called **second-rank tensors**. We will show later that strain is also described as a second-rank tensor. Higher-rank tensors will be used to describe how the action of an applied stimulus will produce a change in the physical properties of a crystal. The susceptibility tensor, eqn 7.4, is a simple example.

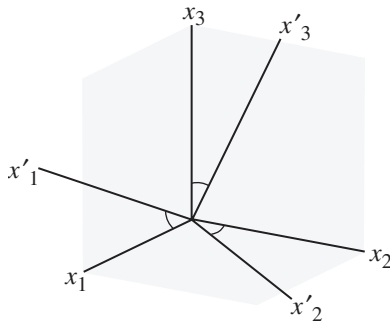


Fig. 7.2 Two sets of orthogonal axes oriented with respect to each other, as used in the transformation equations.

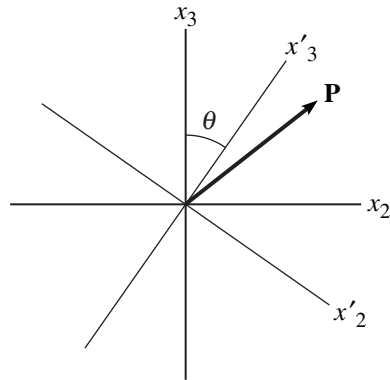


Fig. 7.3 Schematic rotation of axes in two dimensions. θ is the angle between the x_3 and x'_3 axes.

7.1.3 Field and matter tensors

Implicit in the discussion so far is the fact that some tensor quantities, such as the applied field, are controlled by the person setting up an experiment and are independent of the crystal. It is convenient to treat these tensors as special quantities, which are called **field tensors**. On the other hand, there are some tensor quantities that are properties of the material, such as the coefficients of thermal expansion and dielectric susceptibility, and these are called **matter tensors**. In the formalism it often appears as if the matter tensors are constants of the material. However, we will meet examples of how the values of the matter tensors can be modified by application of a field tensor quantity (Section 7.6). This will be particularly noticeable if the application of an external field changes the symmetry of the crystal, because the form of the matter tensors are determined by symmetry.

7.2 First-rank tensors

A first-rank tensor transforms as a vector. Consider the two sets of axes shown in Fig. 7.2, where for the convenience of notation later we label the x, y, z axes of one set of axes as x_1, x_2, x_3 , and of the second set of axes as x'_1, x'_2, x'_3 . We define the matrix of direction cosines as $a_{ij} = \cos(x'_i : x_j)$. This is more easily pictured for the two-dimensional case shown in Fig. 7.3. We define $\theta = x'_3 : x_3 = x'_2 : x_2$, so that $x'_2 : x_3 = 90^\circ + \theta$ and $x'_3 : x_2 = 90^\circ - \theta$. Thus the **matrix of direction cosines** can be written as

$$\mathbf{a} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{pmatrix} \quad (7.5)$$

Note that $a_{ij} = -a_{ji}$ for $i \neq j$, which will be true in general for rotation matrices. It is left as a problem at the end of the chapter (Problem 7.1) to show that for the general case $\mathbf{a}^{-1} = \mathbf{a}^T$; eqn 7.5 is clearly consistent with this.

If we have a vector $\mathbf{P}$ in the first set of axes, this is defined as $\mathbf{P}'$ in the second set of axes, and we have the normal vector transformations:

$$\begin{aligned} \mathbf{P}' &= \mathbf{a} \cdot \mathbf{P}; & \mathbf{P} &= \mathbf{a}^{-1} \cdot \mathbf{P}' = \mathbf{a}^T \cdot \mathbf{P}' \\ \Rightarrow P'_i &= \sum_j a_{ij} P_j; & P_i &= \sum_j a_{ji} P'_j \end{aligned} \quad (7.6)$$

We have already met two first-rank tensors, namely the applied electric field $\mathbf{E}$ and the polarization $\mathbf{P}$. We also have the magnetic analogues $\mathbf{B}$ and $\mathbf{M}$ respectively. Other first-rank tensors are heat flow, temperature gradient, and electric current.

7.3 Second-rank tensors

7.3.1 Basic ideas

We met the susceptibility tensor as a second-rank tensor in eqn 7.4. In fact whenever one quantity described by a first-rank tensor depends on another

first-rank tensor quantity, such as when $\mathbf{P}$ is obtained by application of an external $\mathbf{E}$, the link between the two is described by a second-rank tensor. Before we explore the inner wonders of second-rank tensors in more detail, we deal first with the methods used to rotate a second-rank tensor. We start by considering the following tensor product involving $\mathbf{P}$ and $\mathbf{Q}$ as first-rank tensors and $\mathbf{T}$ as a second-rank tensor:

$$\mathbf{P} = \mathbf{T} \cdot \mathbf{Q} \Rightarrow P_i = \sum_j T_{ij} Q_j \quad (7.7)$$

If we have the same tensors in a different orientation of the axes, we have a corresponding equation, $\mathbf{P}' = \mathbf{T}' \cdot \mathbf{Q}'$. From eqn 7.6, $\mathbf{P}' = \mathbf{a} \cdot \mathbf{P}$ and $\mathbf{Q}' = \mathbf{a} \cdot \mathbf{Q}$. The equation that links $\mathbf{T}'$ to $\mathbf{T}$ is obtained from

$$\begin{aligned} \mathbf{P}' &= \mathbf{a} \cdot \mathbf{P} = \mathbf{a} \cdot (\mathbf{T} \cdot \mathbf{Q}) \\ \mathbf{P}' &= \mathbf{T}' \cdot \mathbf{Q}' = \mathbf{T}' \cdot (\mathbf{a} \cdot \mathbf{Q}) \\ \Rightarrow \mathbf{a} \cdot \mathbf{T} &= \mathbf{T}' \cdot \mathbf{a} \\ \Rightarrow \mathbf{T}' &= \mathbf{a} \cdot \mathbf{T} \cdot \mathbf{a}^T \end{aligned} \quad (7.8)$$

Written out long-hand, this gives

$$T'_{ij} = \sum_{k,l} a_{ik} a_{jl} T_{kl} \quad (7.9)$$

The reverse transformation gives

$$\begin{aligned} \mathbf{T} &= \mathbf{a}^T \cdot \mathbf{T}' \cdot \mathbf{a} \\ \Rightarrow T_{ij} &= \sum_{k,l} a_{ki} a_{lj} T'_{kl} \end{aligned} \quad (7.10)$$

As an example, consider the tetragonal ferroelectric phase of BaTiO_3 . The susceptibility tensor at room temperature has $\chi_{11} = \chi_{22} = 70$, $\chi_{33} = 250$ (see Fig. 2.31). Suppose a crystal is cut in the form of a plate such that the normal to the plate is in the x - z plane at an angle of θ to $[001]$, as shown in Fig. 7.4, and suppose that an electric field $\mathbf{E}'$ is applied normal to the plate. We can determine the resultant polarization field $\mathbf{P}'$ in either of two ways. The first is to transform $\mathbf{E}'$ into the frame of the principal axes by the following transformation:

$$\mathbf{E} = \begin{pmatrix} a_{11}E'_1 + a_{13}E'_3 \\ E'_2 \\ a_{31}E'_1 + a_{33}E'_3 \end{pmatrix} \quad (7.11)$$

We can then obtain $\mathbf{P} = \epsilon_0 \boldsymbol{\chi} \cdot \mathbf{E}$ directly, and then calculate the polarization in the axes of the crystal plate, $\mathbf{P}'$, by the reverse transformation. However, we could do the whole calculation in the axes of the crystal plate by transforming $\boldsymbol{\chi}$. Application of the transformation equation gives

$$\boldsymbol{\chi}' = \begin{pmatrix} a_{11}^2 \chi_{11} + a_{13}^2 \chi_{33} & 0 & a_{11} a_{31} \chi_{11} + a_{13} a_{33} \chi_{33} \\ 0 & \chi_{22} & 0 \\ a_{31} a_{11} \chi_{11} + a_{33} a_{13} \chi_{33} & 0 & a_{31}^2 \chi_{11} + a_{33}^2 \chi_{33} \end{pmatrix} \quad (7.12)$$

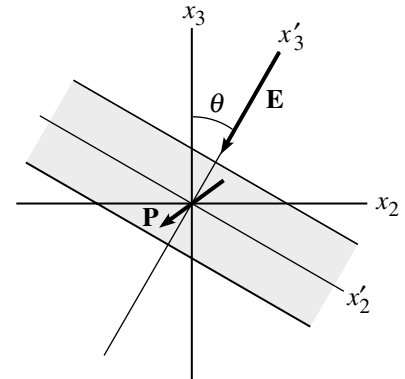


Fig. 7.4 Schematic illustration of an anisotropic dielectric plate, showing how the orientation of the induced $\mathbf{P}$ is not parallel to the applied $\mathbf{E}$ if x_2 and x_3 are the principal crystal axes.

Since the applied electric field is of the form

$$\mathbf{E}' = \begin{pmatrix} 0 \\ 0 \\ E'_3 \end{pmatrix} \quad (7.13)$$

we have

$$\mathbf{P}' = \epsilon_0 \chi' \cdot \mathbf{E}' = \epsilon_0 \begin{pmatrix} (a_{11}a_{31}\chi_{11} + a_{13}a_{33}\chi_{33})E'_3 \\ 0 \\ (a_{31}^2\chi_{11} + a_{33}^2\chi_{33})E'_3 \end{pmatrix} \quad (7.14)$$

Clearly there is now a component of the polarization that is normal to the direction of the applied electric field (Problem 7.11).

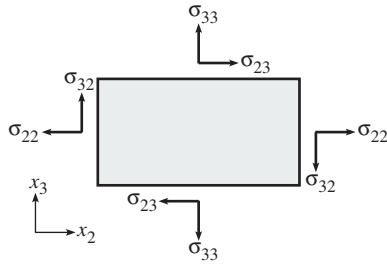


Fig. 7.5 Schematic definition of the full stress tensor. The arrows indicate the direction of the applied force relative to the orientation of the surface.

7.3.2 Stress as a second-rank tensor

Stress arises from applying a force to a surface, and its units are force/area. Force is a vector quantity, and hence is a first-rank tensor. But when you apply force to a surface you have to consider also the relative orientations of the force vector and the normal to the surface. This is illustrated in Fig. 7.5. In the case of an isotropic material, if the force is applied normal to the surface, the result will be a compression of the crystal in the direction of the force. But if the force is applied in a direction parallel to the face of the crystal, the result is a shear of the crystal. To distinguish between these two outcomes, we need a definition of stress that includes both the direction of the force and the direction of the normal to the face the force is acting on. If the two directions are parallel, we have a tensile stress, and if the two directions are orthogonal we have a shear stress. When a force is applied at an arbitrary direction to a surface, we have a mixture of the two types of stress, which can be obtained by separating the force vector into the components normal and parallel to the face. The **stress tensor** is given the symbol σ .

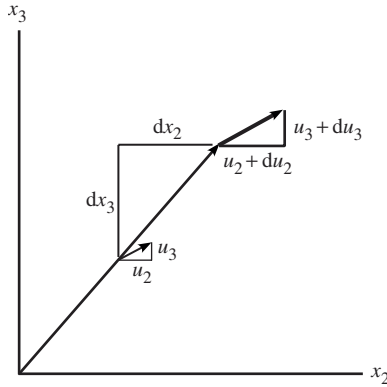


Fig. 7.6 Schematic two-dimensional representation of how a displacement vector (u_2, u_3) for a small element of a body under strain deformation will vary depending on position (x_2, x_3) . How the changes in displacements (du_2, du_3) depend on (x_2, x_3) is described by the strain tensor ϵ .

7.3.3 Strain as a second-rank tensor

If stress is a second-rank tensor, we might expect the response, namely the strain, also to be a second-rank tensor. We noted before that strain includes effects on both the size and shape of an object: formally this is both the tensile and shear strain.

Strain is formally defined by the relationship between the displacement of a small element of a solid and its position. This is illustrated for two dimensions in Fig. 7.6, which shows how a general displacement induced by the strain varies according to position. The tensile strain relates the displacement along one direction to the position in the same direction. A small element of the material at initial position x_2, x_3 will be displaced by u_2, u_3 , and the element at position $x_2 + dx_2, x_3 + dx_3$ will be displaced by $u_2 + du_2, u_3 + du_3$. The dependence of u_2 on x_2 , x_3 respectively is described by the general equation for tensile strain:

$$e_{ii} = \frac{du_i}{dx_i} \quad (7.15)$$

Now we consider the shear strain, which links the dependence of u_2 on x_3 (and similarly for other combinations). We write the shear strain as

$$e_{23} = \frac{du_2}{dx_3} \quad (7.16)$$

Clearly we can establish a general definition that encompasses both types of strain:

$$e_{ij} = \frac{du_i}{dx_j} \quad (7.17)$$

There is, however, an additional complication with the shear strains that did not arise with the definition of stress. The problem with the general definition is that a pure rotation of an object will also give rise to a value of e_{ij} . This is illustrated in Fig. 7.7. For a general deformation we have different magnitudes of e_{ij} and e_{ji} , representing a mixture of shear strain and rotation. This problem is resolved by defining a new strain tensor ϵ as the symmetric part of the tensor $\mathbf{e}$:

$$\epsilon_{ij} = \frac{1}{2}(e_{ij} + e_{ji}) \quad (7.18)$$

and a rotation tensor ω as the antisymmetric part of $\mathbf{e}$:

$$\omega_{ij} = \frac{1}{2}(e_{ij} - e_{ji}) = -\omega_{ji} \quad (7.19)$$

These two combinations are illustrated in Fig. 7.7. The rotation matrix ω will not interest us further; the pure shear strain tensor ϵ contains all the important information.

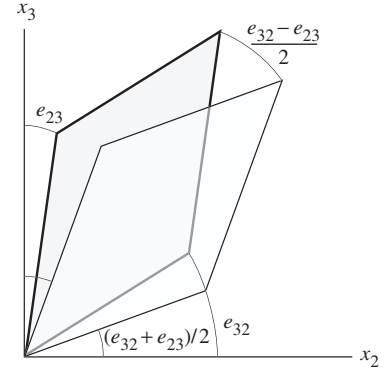


Fig. 7.7 Schematic representation of the relationship between shear strain and rotation. The initial object is a rectangle with edges parallel to the axes. $(e_{32} + e_{23})/2$ defines the shear strain, and $(e_{32} - e_{23})/2$ defines the rotation.

7.3.4 45° rotation of the strain tensor and the conversion between tensile and shear strain

The fact that we can link both tensile and shear strain into a single tensor suggests that the distinctions of the two types of strain may depend on the definition of the coordinate system. This point is illustrated for a 45° rotation of the coordinate system about the x_1 -axis, as shown in Fig. 7.8. We have the following transformation matrix:

$$\mathbf{a} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & -1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix} \quad (7.20)$$

A tensor that describes a pure shear strain in the x_2 - x_3 plane is

$$\epsilon = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & \epsilon \\ 0 & \epsilon & 0 \end{pmatrix} \quad (7.21)$$

When we transform the axes, we obtain the new strain tensor:

$$\epsilon'_{22} = a_{22}a_{23}\epsilon_{23} + a_{23}a_{22}\epsilon_{32} \quad (7.22)$$

$$\epsilon'_{33} = a_{32}a_{33}\epsilon_{23} + a_{33}a_{32}\epsilon_{32} = -\epsilon'_{22} \quad (7.23)$$

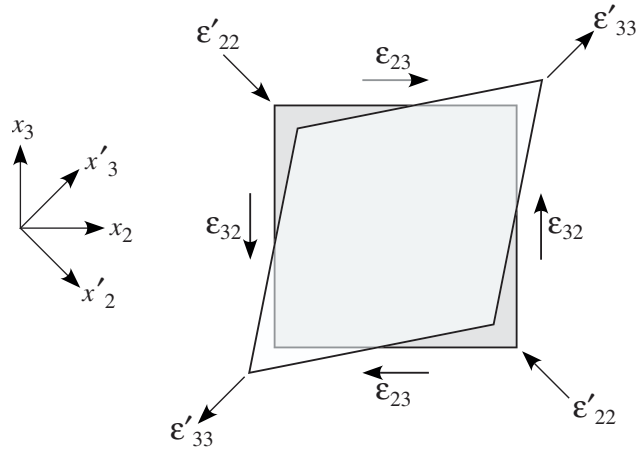


Fig. 7.8 Rotation of a shear strain tensor ϵ leads to a tensile strain tensor ϵ' .

with all other components equal to zero. Thus

$$\epsilon' = \begin{pmatrix} 0 & 0 & 0 \\ 0 & \epsilon & 0 \\ 0 & 0 & -\epsilon \end{pmatrix} \quad (7.24)$$

What is clear from this result is that the rotation of the axes by 45° has led to the conversion of the definition of the type of strain from pure shear to pure tensile.

We can also remark that if the initial crystal has volume V , the new volume following application of the strain tensor described by eqn 7.24 will be

$$V(1 + \epsilon)(1 - \epsilon) = V(1 - \epsilon^2).$$

This demonstrates that a pure shear strain gives only a quadratic change in volume.

7.3.5 Voigt notation

It will prove to be useful to simplify the notation used in the indices. In what is called the **Voigt notation**, the pairs of indices 11, 22, 33 are replaced by the single indices 1, 2, and 3 respectively, and the pairs of indices 23, 13, and 12 are replaced by the single indices 4, 5, and 6 respectively. Thus the strain tensor can be represented by

$$\epsilon = \begin{pmatrix} \epsilon_{11} & \epsilon_{12} & \epsilon_{13} \\ \epsilon_{21} & \epsilon_{22} & \epsilon_{23} \\ \epsilon_{31} & \epsilon_{32} & \epsilon_{33} \end{pmatrix} = \begin{pmatrix} \epsilon_1 & \epsilon_6 & \epsilon_5 \\ \epsilon_6 & \epsilon_2 & \epsilon_4 \\ \epsilon_5 & \epsilon_4 & \epsilon_3 \end{pmatrix} \equiv \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \\ \epsilon_4 \\ \epsilon_5 \\ \epsilon_6 \end{pmatrix} \quad (7.25)$$

This notation makes the labelling easier, and makes the representation of higher-rank tensors considerably easier. For example, the tensor linking strain and

stress is the fourth-rank **elastic constant tensor**. The Voigt notation reduces the number of elements from $3^4 = 81$ to $6 \times 6 = 36$. We will shortly meet some complicating rules that are implied by the use of the Voigt notation (Section 7.4.2), and we have to concede that it is much easier not to use the Voigt notation when rotating tensors (Section 7.4.3), but the ease of representation has led to the near-universal use of the Voigt notation.

7.3.6 Principal axes

Implicit in what we have said so far is that the second-rank matter tensors are symmetric. This arises from the fact that there is for any crystal a set of orthogonal axes for which a matter tensor will be diagonal. These are called the **principal axes**. A vector field applied along one of the principal axes will produce a vector field along the same direction. In tetragonal, hexagonal, and trigonal crystals, one of the principal axes is along [001] (the perpendicular plane is isotropic, so any orientation of the other two axes is equivalent to any other). In orthorhombic systems the principal axes are parallel to the three lattice vectors. For monoclinic crystals one of the principal axes is parallel to [010], with the other two axes lying somewhere in the **a–c** plane.

Now we consider the rotation of a diagonal second-rank tensor. We take the following two components:

$$T'_{ij} = \sum_{kl} a_{ik} a_{jl} T_{kl} = \sum_k a_{ik} a_{jk} T_{kk} \quad (7.26)$$

$$T'_{ji} = \sum_{kl} a_{jk} a_{il} T_{kl} = \sum_k a_{jk} a_{ik} T_{kk} \quad (7.27)$$

Clearly $T'_{ij} = T'_{ji}$, which demonstrates that the second-rank matter tensor will be symmetric for all orientations of the axes.

It is useful to be able to obtain the orientation of the principal axes of a second-rank tensor. We illustrate the rotation involved with an example that will require rotation about only one axis:

$$\mathbf{T} = \begin{pmatrix} T_{11} & 0 & T_{13} \\ 0 & T_{22} & 0 \\ T_{31} & 0 & T_{33} \end{pmatrix} \quad (7.28)$$

To reduce this tensor to its principal axes requires rotation about y. The transformation matrix is

$$\mathbf{a} = \begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix} \quad (7.29)$$

When the tensor has been transformed to its principal axes, $T'_{13} = 0$. This allows us to write (noting that $T_{31} = T_{13}$)

$$\begin{aligned} T'_{13} = 0 &= a_{11}a_{31}T_{11} + a_{11}a_{33}T_{13} + a_{13}a_{31}T_{31} + a_{13}a_{33}T_{33} \\ &= \cos \theta \sin \theta (T_{33} - T_{11}) + (\cos^2 \theta - \sin^2 \theta) T_{13} \\ &= \frac{1}{2} \sin 2\theta (T_{33} - T_{11}) + \cos 2\theta T_{13} \\ \Rightarrow \tan 2\theta &= \frac{2T_{13}}{T_{11} - T_{33}} \end{aligned} \quad (7.30)$$

From this equation, the rotation angle θ can be obtained, from which the full transformation matrix and hence the principal components of the tensor can be determined. In the case where we need to find three angles, the geometric method is not practical. In this case, the principal components are equivalent to the eigenvalues of the second-rank tensor, and the transformation matrix is the corresponding set of eigenvectors.

7.3.7 Symmetry and second-rank matter tensors

It is straightforward to appreciate that symmetry will lead to some components of second-rank matter tensors being equal when in the setting of the principal axes. For example, for cubic symmetry the three diagonal components will be equal. For tetragonal, hexagonal, and trigonal symmetry, the matter tensor components have $T_{11} = T_{22} \neq T_{33}$.

Symmetry plays an important role in determining whether some of the components of a matter tensor should be zero. We consider two examples. The first is the case of a 2-fold rotation axis along y . The relationship between atomic coordinate $x, y, z \rightarrow \bar{x}, y, \bar{z}$ has the corresponding transformation matrix:

$$\mathbf{a} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \quad (7.31)$$

The symmetry requires that the transformed tensor $\mathbf{T}'$ must be identical to the initial tensor $\mathbf{T}$. The transformed tensor is obtained as

$$\mathbf{T}' = \begin{pmatrix} T_{11} & -T_{12} & T_{13} \\ -T_{21} & T_{22} & -T_{23} \\ T_{31} & -T_{23} & T_{33} \end{pmatrix} \quad (7.32)$$

The solution to $T'_{12} = T_{12} = -T_{12}$ is $T_{12} = 0$. Similarly, $T_{23} = 0$. On the other hand, since $T'_{13} = T_{13}$, this component of the tensor can have a non-zero value. The same result will follow for the transformation of a mirror plane normal to $[010]$. This simple analysis shows how a symmetry transformation can be used to determine which coefficients of a matter tensor must have a value of zero and which are allowed to have non-zero values.

7.3.8 Example of zero thermal expansion

We now return to our initial example of a material that has both positive and negative values of the coefficient of the thermal expansion tensor. Since thermal expansion involves the strain tensor, the thermal expansion tensor α is second-rank, and for a scalar change in temperature ΔT we have

$$\epsilon = \alpha \cdot \Delta T \quad (7.33)$$

Now we consider a uniaxial material in which $\alpha_{11} = \alpha_{22} < 0$ and $\alpha_{33} > 0$, the example shown in Fig. 7.1. If we define the direction normal to the plate to be x'_3 , we have

$$\alpha'_{33} = a_{32}^2 \alpha_{22} + a_{33}^2 \alpha_{33} = 0 \quad (7.34)$$

We now need to find a value for the rotation angle such that $\alpha'_{33} = 0$. Substituting the trigonometric values of the rotation matrix elements a_{ij} , we obtain

$$\tan^2 \theta = -\alpha_{33}/\alpha_{22} \quad (7.35)$$

Now we consider the other strain components. These are given as

$$\alpha'_{22} = a_{22}^2 \alpha_{22} + a_{23}^2 \alpha_{33} \quad (7.36)$$

$$\alpha'_{23} = a_{22} a_{32} \alpha_{22} + a_{23} a_{33} \alpha_{33} \quad (7.37)$$

Not only is there an expansion of the plate in the plane of the plate, but there is also a shear strain in the plane of the plate induced by the temperature change.

7.4 Third-rank tensors

7.4.1 Piezoelectricity

The definition of the piezoelectric tensor

Arguably one of the most important third-rank tensors for 21st-century applications is the **piezoelectric tensor**, d_{ijk} . This links an induced polarization to an applied stress (the **direct** piezoelectric effect), or an induced strain to an applied electric field (the **indirect** piezoelectric effect):

$$P_i = \sum_{jk} d_{ijk} \sigma_{jk} \quad (7.38)$$

$$\epsilon_{jk} = \sum_i d_{ijk} E_i \quad (7.39)$$

The equivalence of these forms can be obtained by a thermodynamic argument. We write the internal energy as

$$dU = \sum_{jk} \sigma_{jk} d\epsilon_{jk} + \sum_i E_i dP_i \quad (7.40)$$

The free energy function, which we denote as Φ , can be written as

$$\Phi = U - \sum_{jk} \sigma_{jk} \epsilon_{jk} - \sum_i E_i P_i \quad (7.41)$$

This can be written in differential form as

$$d\Phi = - \sum_{jk} \epsilon_{jk} d\sigma_{jk} - \sum_i P_i dE_i \quad (7.42)$$

Thus we have the following two differentials:

$$\left. \frac{\partial \Phi}{\partial \sigma_{jk}} \right|_E = -\epsilon_{jk} \quad (7.43)$$

$$\left. \frac{\partial \Phi}{\partial E_i} \right|_\sigma = -P_i \quad (7.44)$$

From the second differential we have

$$-\frac{\partial^2 \Phi}{\partial E_i \partial \sigma_{jk}} = \frac{\partial \epsilon_{jk}}{\partial E_i} = \frac{\partial P_i}{\partial \sigma_{jk}} = d_{ijk} \quad (7.45)$$

Both first differentials relate to the piezoelectric effect, as the indirect and direct effect respectively. This demonstrates that the same coefficient applies to both effects.

Symmetry and piezoelectricity: example of the tetragonal ferroelectric phase of perovskite

Symmetry plays an important role in determining the piezoelectric properties of a material. Consider the direct piezoelectric effect. The simplest example is the role of a centre of symmetry. An applied stress will not remove the centre (stress always has a symmetry corresponding to a centrosymmetric point group), and so application of stress by itself cannot create an electrical polarization in a centrosymmetric crystal. Therefore the first symmetry condition that must be met for a piezoelectric material is that it will not have a centre of symmetry. There are further symmetry conditions on the type of stress that can induce an electrical polarization, and on the direction of that polarization.

The role of symmetry can be illustrated by the example of the piezoelectric effect given by the ferroelectric tetragonal phase of a perovskite material such as BaTiO_3 or PbTiO_3 , which is shown schematically in Fig. 7.9. The symmetry of the crystal is point group $4mm$, that is, there is a 4-fold rotation axis along $[001]$, and mirror planes perpendicular to the equivalent $\langle 100 \rangle$ and $\langle 110 \rangle$ directions. Because of the symmetry of the ferroelectric phase there is already an electrical polarization along $[001]$ in the unstressed crystal, together with an elongation of the crystal in this direction. Application of the stress σ_3 , that is a force along $[001]$ applied on the (001) face, will reduce the extent of the elongation along the $[001]$ direction. This will squash the TiO_6 octahedra and force the Ti^{4+} cations back towards the centre of the octahedra. Similarly, there will be a compression of the oxygen atoms around the Ba/Pb sites, forcing the cation to move towards the centre of the 12 oxygens. The effects of these induced cation displacements will be to reduce the overall electrical polarization along $x_3 = [001]$.

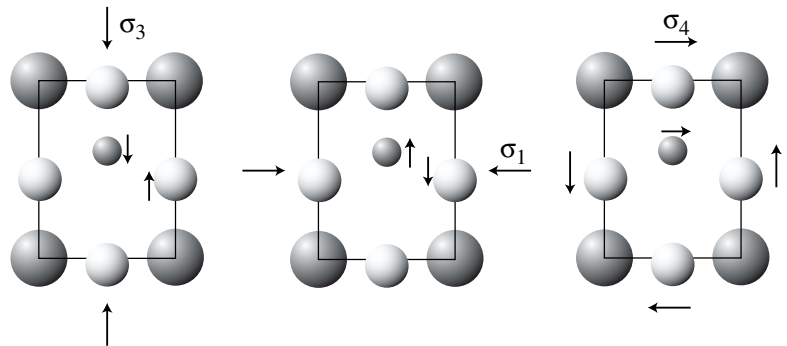


Fig. 7.9 Schematic representation of the piezoelectric effect in a ferroelectric perovskite of point group symmetry $4mm$. $[001]$ (x_3) is vertical, and $[100]$ (x_1) is horizontal.

Now consider application of the stress σ_1 (because of the 4-fold rotational symmetry the same arguments here will also apply to σ_2). There is a mirror plane normal to the face on which the force is applied, and the stress is not able to change this symmetry. Therefore there will be no induced electrical polarization along the $[100]$ direction. However, by squashing together the oxygen atoms in the TiO_6 octahedra, the Ti^{4+} cation will be further displaced along $[001]$, and so an additional contribution to the electrical polarization along $[001]$ will be induced.

We now consider the effects of an applied shear stress. First we consider σ_6 , which will cause a shear of the **a–b** plane (not illustrated in Fig. 7.9). This will have the effect of moving two of the oxygens in the TiO_6 octahedra closer together, and of moving two others further apart. This shear does not destroy the orthogonal mirror planes on the $\{110\}$ planes, so there can be no polarization in the **a–b** plane. To first order, the changes in the distances between the two pairs of oxygen atoms in the **a–b** plane along the $\langle 110 \rangle$ directions will be of equal size but opposite sign, so there will no net effect on the position of the Ti^{4+} cation to first order (although there may be a second-order effect that would be represented by a tensor of higher order). Thus σ_6 is not able to produce an induced electrical polarization. On the other hand σ_5 , which causes a shear on the **a–c** plane, removes all mirror planes other than the (010) mirror plane, and therefore a polarization is allowed on a general direction in the **a–c** plane.

Writing everything out full, we can state the results of this discussion in matrix form:

$$\begin{pmatrix} P_1 \\ P_2 \\ \Delta P_3 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 & d_{15} & 0 \\ 0 & 0 & 0 & d_{15} & 0 & 0 \\ d_{31} & d_{31} & d_{33} & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_3 \\ \sigma_4 \\ \sigma_5 \\ \sigma_6 \end{pmatrix} \quad (7.46)$$

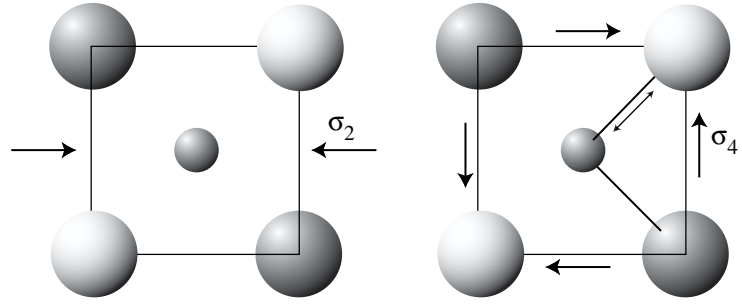
where we have noted that $d_{32} = d_{31}$ and $d_{24} = d_{15}$, and we have explicitly noted that P_1 and P_2 are new, but that P_3 changes by ΔP_3 .

Piezoelectricity of non-polar systems: example of a tetrahedron

The example of the tetragonal ferroelectric phase of perovskite is a case where the existing polarization of the crystal is changed by the application of stress. Piezoelectricity is also found in crystals without a centre of symmetry even when there is no polarization in the stress-free crystal. We explore this point by the example of a single tetrahedral group of ions, as illustrated in Fig. 7.10. In this case we assume that the vertices of the tetrahedra are anions, of charge $-Q$, and the centre of the cluster is a cation, of charge $+4Q$. Note that the point symmetry of the tetrahedral cluster is $\bar{4}3m$ – there is no centre of symmetry. We will assume for the sake of the discussion that the anion–cation bond is strong, and remains of constant length when a small stress is applied. This can be achieved if the anion–cation–anion bonds can flex with the applied stress to give an angle different to the ideal tetrahedral angle of 109.47° . In Fig. 7.10 the axes are defined along the $\bar{4}$ axes, with the central cation at the origin.

We first consider the application of the stress σ_1 . This has the effect of squashing the tetrahedral cluster along the x_1 direction. This causes the 3-fold

Fig. 7.10 Schematic representation of the piezoelectric effect in a tetrahedron of initial point group symmetry $\bar{4}3m$. The x_1 axis is horizontal, and the x_2 axis is vertical. The lightly-shaded anions (large spheres) are above the plane of the diagram, and the darkly-shaded anions are below the plane of the diagram; the central cation (small sphere) is in the plane of the diagram.



rotational symmetry to be lost along each of the axes, and the $\bar{4}$ rotoinversion axes along both the x_2 and x_3 axes are reduced to 2-fold rotation axes along these directions. However, the $\bar{4}$ rotoinversion axis along x_1 is retained. With the presence of these three orthogonal rotation and rotoinversion axes, there can be no polarization.

Now we consider the application of a shear stress, σ_6 . This has the effect of moving the top two anions away from each other, and the lower two anions apart. Since the system tries to keep the cation–anion distances constant, the cation is moved upwards. It can be shown (Problem 7.4) that the displacement of the cation along the x_3 direction, $u_{(z)}^+$, is related to the displacements of the anions in the $x_1 - x_2(x - y)$ direction, $\pm u_{(xy)}^-$, by

$$|u_{(z)}^+| = |u_{(xy)}^-|/\sqrt{2} \quad (7.47)$$

where we use the modulus because the different directions of motion all cause the same displacement. The induced polarization is simply equal to $4Qu_{(z)}^+$.

The final result can be summarized by the matrix equation:

$$\begin{pmatrix} P_1 \\ P_2 \\ P_3 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & d_{13} & 0 & 0 \\ 0 & 0 & 0 & 0 & d_{13} & 0 \\ 0 & 0 & 0 & 0 & 0 & d_{13} \end{pmatrix} \begin{pmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_3 \\ \sigma_4 \\ \sigma_5 \\ \sigma_6 \end{pmatrix} \quad (7.48)$$

where we have used the fact that $d_{13} = d_{25} = d_{36}$ by symmetry.

Given that a perfect tetrahedral arrangement of cations and anions has a non-zero piezoelectric tensor, we might expect crystals that contain tetrahedral arrangements of atoms to also have non-zero piezoelectric tensor. This applies to the semiconducting cubic ZnS and related materials, where each cation is in a tetrahedral coordination with neighbouring anions (and *vice versa*) and to systems such as quartz that consist of networks of linked SiO_4 tetrahedra. Of course, the arrangements of the tetrahedral groups may give additional symmetry that changes the piezoelectric properties – for example, the crystal structure of the high-temperature cristobalite phase of silica has a centre of symmetry – but the point is that a crystal without a centre of symmetry can have

a piezoelectric effect even without there being a polarization in the stress-free crystal.

7.4.2 Use of Voigt notation for third-rank tensors

We have made use of the Voigt notation in the example above, but to perform operations such as the rotation of axes we need to convert to the long-handed three-index mode. In long hand, we can write the equation for the direct piezoelectric effect as

$$\begin{aligned} P_1 = & d_{111}\sigma_{11} + d_{112}\sigma_{12} + d_{113}\sigma_{13} \\ & + d_{121}\sigma_{21} + d_{122}\sigma_{22} + d_{123}\sigma_{23} \\ & + d_{131}\sigma_{31} + d_{132}\sigma_{32} + d_{133}\sigma_{33} \end{aligned}$$

and similarly for P_2 and P_3 . When we take account of the symmetry of the σ tensor by using Voigt notation for this tensor, i.e. $\sigma_{12} = \sigma_{21} = \sigma_6$, etc., we can rewrite this equation as

$$\begin{aligned} P_1 = & d_{111}\sigma_1 + (d_{112} + d_{121})\sigma_6 + (d_{113} + d_{131})\sigma_5 \\ & + d_{122}\sigma_2 + (d_{123} + d_{132})\sigma_4 + d_{133}\sigma_3 \end{aligned}$$

By comparison with the full Voigt expansion:

$$P_1 = d_{11}\sigma_1 + d_{12}\sigma_2 + d_{13}\sigma_3 + d_{14}\sigma_4 + d_{15}\sigma_5 + d_{16}\sigma_6 \quad (7.49)$$

we see that $d_{11} = d_{111}$, $d_{12} = d_{122}$, and $d_{13} = d_{133}$, whereas $d_{14} = 2d_{123}$, $d_{15} = 2d_{113}$, and $d_{16} = 2d_{112}$. Similar results follow from an analysis of P_2 and P_3 . The new factor of 2 is important!

7.4.3 Transformations of third-rank tensors

Transformation equations

Earlier we used physical arguments to obtain the non-zero components of the piezoelectric tensor for a tetragonal crystal and a tetrahedron. The symmetry properties can be obtained by using symmetry transformations, but for this we need to see how to transform a third-rank tensor by a change in the definitions of the axes. This will follow the approach taken in transformation of second-rank tensors. We define a first-rank tensor $\mathbf{P}$, a second-rank tensor $\mathbf{Q}$, and a third-rank tensor $\mathbf{T}$. These are linked by $\mathbf{P} = \mathbf{T} \cdot \mathbf{Q}$. Similarly we can define a set of transformed tensors: $\mathbf{P}' = \mathbf{T}' \cdot \mathbf{Q}'$. Following the earlier procedure, we obtain

$$\begin{aligned} \mathbf{P}' &= \mathbf{a} \cdot \mathbf{P} = \mathbf{a} \cdot (\mathbf{T} \cdot \mathbf{Q}) \\ \mathbf{P}' &= \mathbf{T}' \cdot \mathbf{Q}' = \mathbf{T}' \cdot (\mathbf{a} \cdot \mathbf{Q} \cdot \mathbf{a}^T) \\ \Rightarrow \mathbf{a} \cdot \mathbf{T} \cdot \mathbf{Q} &= \mathbf{T}' \cdot \mathbf{a} \cdot \mathbf{Q} \cdot \mathbf{a}^T = \mathbf{T}' \cdot \mathbf{a} \cdot \mathbf{a} \cdot \mathbf{Q} \\ \Rightarrow \mathbf{T}' &= \mathbf{a} \cdot \mathbf{T} \cdot \mathbf{a}^T \cdot \mathbf{a}^T = \mathbf{a} \cdot \mathbf{a} \cdot \mathbf{a} \cdot \mathbf{T} \end{aligned} \quad (7.50)$$

where we have made use of the fact that $\mathbf{Q}^T = \mathbf{Q}$ and $\mathbf{T}^T = \mathbf{T}$, for example in writing $\mathbf{Q} \cdot \mathbf{a}^T = \mathbf{a} \cdot \mathbf{Q}$. In long hand, this can be written as

$$T'_{ijk} = \sum_{lmn} a_{il} a_{jm} a_{kn} T_{lmn} \quad (7.51)$$

with the reverse transformation

$$T_{ijk} = \sum_{lmn} a_{li} a_{mj} a_{nk} T'_{lmn} \quad (7.52)$$

These equations can be readily extended for higher-rank tensor transformations.

Effect of symmetry

We now use the transformation equation to demonstrate how symmetry can determine which elements of the piezoelectric tensor are zero. We will consider the effect of 4-fold rotation axis along x_3 . The transformation matrix for a 90° rotation is

$$\mathbf{a} = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (7.53)$$

First we consider d_{11} and d_{22} . The transformation equations give

$$\left. \begin{aligned} d'_{111} &= a_{12} a_{12} a_{12} d_{222} = d_{222} \\ d'_{222} &= a_{21} a_{21} a_{21} d_{111} = -d_{111} \end{aligned} \right\} \Rightarrow d_{11} = -d_{22} = 0$$

For d_{33} we have

$$d'_{333} = a_{33} a_{33} a_{33} d_{333} = d_{333} \Rightarrow d_{33} \neq 0$$

We now consider the pairs involving the tensile stresses, using the fact that we expect pairs to be equal by symmetry:

$$\left. \begin{aligned} d'_{122} &= a_{12} a_{21} a_{21} d_{211} = d_{211} \\ d'_{211} &= a_{21} a_{12} a_{12} d_{112} = -d_{112} \end{aligned} \right\} \Rightarrow d_{12} = -d_{21} = 0$$

$$\left. \begin{aligned} d'_{133} &= a_{12} a_{33} a_{33} d_{233} = d_{233} \\ d'_{233} &= a_{21} a_{33} a_{33} d_{133} = -d_{133} \end{aligned} \right\} \Rightarrow d_{13} = -d_{23} = 0$$

$$\left. \begin{aligned} d'_{311} &= a_{33} a_{12} a_{12} d_{322} = d_{322} \\ d'_{322} &= a_{33} a_{21} a_{21} d_{311} = d_{311} \end{aligned} \right\} \Rightarrow d_{31} = d_{32} \neq 0$$

Now we consider the piezoelectric coefficients involving the shear stresses:

$$\left. \begin{aligned} d'_{123} &= a_{12} a_{21} a_{33} d_{213} = -d_{213} \\ d'_{213} &= a_{21} a_{12} a_{33} d_{123} = -d_{123} \end{aligned} \right\} \Rightarrow d_{14} = -d_{25} \neq 0$$

$$\left. \begin{aligned} d'_{223} &= a_{21} a_{21} a_{33} d_{113} = d_{113} \\ d'_{113} &= a_{12} a_{12} a_{33} d_{223} = d_{223} \end{aligned} \right\} \Rightarrow d_{15} = d_{24} \neq 0$$

$$\left. \begin{aligned} d'_{323} &= a_{33} a_{21} a_{33} d_{313} = -d_{313} \\ d'_{313} &= a_{33} a_{12} a_{33} d_{323} = d_{323} \end{aligned} \right\} \Rightarrow d_{35} = -d_{34} = 0 \quad (7.54)$$

$$\left. \begin{aligned} d'_{121} &= a_{12} a_{21} a_{12} d_{212} = -d_{212} \\ d'_{212} &= a_{21} a_{12} a_{21} d_{121} = d_{121} \end{aligned} \right\} \Rightarrow d_{26} = -d_{16} = 0$$

$$d'_{312} = a_{33} a_{12} a_{21} d_{321} = -d_{321} \Rightarrow d_{36} = 0$$

This gives the piezoelectric tensor for a material with point group 4 of

$$\mathbf{d} = \begin{pmatrix} 0 & 0 & 0 & d_{14} & d_{15} & 0 \\ 0 & 0 & 0 & d_{15} & -d_{14} & 0 \\ d_{31} & d_{31} & d_{33} & 0 & 0 & 0 \end{pmatrix} \quad (7.55)$$

As a result of symmetry, the number of non-zero coefficients of the piezoelectric coefficient will be reduced to a relatively small number in most crystal classes.

7.5 Fourth-rank tensors

7.5.1 A hierarchy of higher-order tensors

The principles developed in this chapter, including the methods for transforming tensors when there is a rotation of the frame of reference, and the role of symmetry in determining which coefficients of a matter tensor are allowed to have non-zero values, will apply in the same way (albeit with rather more tedious algebra) to higher-rank tensors. Therefore we will mention only one fourth-rank tensor in closing this chapter, namely the elasticity tensors that link stress and strain.

Since all the field tensors first or are second-rank, the higher-order tensors will correspond to cases where the application of one field tensor is able to change the values of the matter tensors. We will consider briefly examples of this kind in the next section, which will be third-rank and fourth-rank effects.

7.5.2 The elasticity tensors

There are two important tensors that link stress to strain, namely the **elastic compliance tensor** that links an induced strain to an applied stress,

$$\epsilon_{ik} = \sum_{jl} s_{ijkl} \sigma_{jl} \quad (7.56)$$

and the **elastic constant tensor** that gives the value of an applied stress required to produce a given set of values of the strains:

$$\sigma_{ik} = \sum_{jl} c_{ijkl} \epsilon_{jl} \quad (7.57)$$

The energy that accompanies a set of strains is

$$E = \frac{1}{2} \sum_{ijkl} c_{ijkl} \epsilon_{ij} \epsilon_{kl} \quad (7.58)$$

Although there are 81 ($=3^4$) terms in the full elastic constant tensor, the number of independent terms is smaller. Use of the symmetry implicit in the Voigt notation, namely that $c_{ijkl} = c_{jikl} = c_{ijlk} = c_{jilk}$ reduces the number of independent coefficients to $6 \times 6 = 36$. Within the Voigt notation, we also have the symmetry that $c_{mn} = c_{nm}$, so the maximum number of independent

coefficients is 27. Symmetry of the crystal will reduce this further. For example, in the case of a cubic material the elastic constant tensor can be written as

$$\mathbf{c} = \begin{pmatrix} c_{11} & c_{12} & c_{12} & 0 & 0 & 0 \\ c_{12} & c_{11} & c_{12} & 0 & 0 & 0 \\ c_{12} & c_{12} & c_{11} & 0 & 0 & 0 \\ 0 & 0 & 0 & c_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & c_{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & c_{44} \end{pmatrix} \quad (7.59)$$

In an isotropic material, $c_{44} = (c_{11} - c_{12})/2$. Also, in cubic crystals in which all the interatomic interactions can be realistically described by pair potentials, $c_{44} = c_{12}$. In NaCl, $c_{12} = 1.24$ GPa and $c_{44} = 1.26$ GPa. This result holds to a similar or slightly lesser extent in other alkali halide crystals. It is not so appropriate for cubic metals. For example, in Cu, $c_{12} = 12.1$ GPa and $c_{44} = 7.5$ GPa, although in Ni the agreement is rather closer, $c_{12} = 14.0$ GPa and $c_{44} = 12.5$ GPa.

7.6 Induced changes in matter tensors

7.6.1 Basic ideas

When we separated the types of tensors into matter tensors and field tensors, we implied that the field tensors can be varied (by turning a dial on a piece of equipment, say), whereas the matter tensors were constants of the material. However, the values of matter tensors are subject to change if there are changes in the external conditions of the material. For example, changing temperature may give rise to a phase transition, which may have a dramatic effect on the matter tensors, such as the dielectric susceptibility as seen earlier in Figs 1.8 and 2.31. In the case of the ferroelectric perovskites, the high-temperature cubic phase has a centre of symmetry and no non-zero piezoelectric coefficients, but some of these coefficients (e.g. d_{33} , as seen above) will become non-zero on cooling through the phase transition, and further cooling will lead to more distortion of the crystal structure and increases in the sizes of the piezoelectric coefficients. A similar example for quartz will be shown later in Section 12.2.4. Other matter tensors can be changed by application of higher-order field tensors. This is exemplified by the effects of external conditions on the optical properties of materials.

7.6.2 Refractive index, the electro-optic effect, and the photoelastic effect

The refractive index of a material is given as the square root of the high-frequency dielectric constant. In principle, this is also a tensor property. For an anisotropic crystal, the refractive indices along the principle directions will therefore be equal to $n_{ii} = \varepsilon_{ii}^{1/2}$. For many applications, the primary tensor used is $\mathbf{B}$, where the principal coefficients are related to the refractive index by $B_{ii} = 1/n_{ii}^2$.

The tensor $\mathbf{B}$ is a matter tensor, but this can be affected by application of an electric field or elastic stress. Specifically, these effects are written as the

change in the coefficients of the **B** tensor:

$$\Delta B_{ij} = \sum_k z_{ijk} E_k + \sum_{kl} \pi_{ijkl} \sigma_{kl} \quad (7.60)$$

The third-rank tensor **z** is called the **linear electro-optic** tensor, and the fourth-rank tensor **π** is the **piezo-elastic** tensor. Clearly the latter term could also be written using the strain tensor instead.

Finally, the first term in eqn (7.60) describes the linear dependence of the refractive index on applied electric field. There are many cases where the linear dependence is zero, and instead a strong second-order dependence is seen. This is described by a fourth-rank tensor:

$$\Delta B_{ij} = \sum_{kl} \alpha_{ijkl} E_k E_l \quad (7.61)$$

When allowing for higher-order effects, it is clear that these will require higher-rank tensors. The extension of the analysis for these higher-rank tensors follows from the main results presented here.

Summary of chapter

- Physical properties in anisotropic materials are formally described using the language of **tensors**.
- The important feature of tensors is that they can be transformed following a transformation of the system of axes.
- Each crystal has a set of axes – the **principal axes** – for which its properties are described by a tensor in which only the diagonal components have non-zero values.
- The non-zero values for the tensor properties of a crystal are determined by symmetry.

Further reading

The main source texts for tensor properties of crystals are Lovett (1989) and Nye (1985). The mathematics of tensors are described by Sands (1982, 1995).

Exercises

- (7.1) Show that the transformation matrix **a** of eqn 7.5 has the condition $\mathbf{a}^{-1} = \mathbf{a}^T$.
- (7.2) The coefficients of the thermal expansion tensor of calcite (point group $\bar{3}m$) are given in Chapter 1 (page 7). Calculate the orientation of the normal of a plate that can be cut so that it does not change its thickness on heating. Calculate the shear strain induced within the plane if there is a change in temperature of 100 K.
- (7.3) A cubic crystal is cut in the form of a plate parallel to (111). Denoting the axis normal to the plate as x'_3 , if a

stress σ'_3 is applied to the plate, what are the components of the stress tensor described in the reference frame of the principal axes?

- (7.4) Derive eqn 7.47 for the constraint that the bond lengths do not change.
- (7.5) The non-zero components of the piezoelectric tensor for a cubic crystal of point group $\bar{4}3m$ are $d_{14} = d_{25} = d_{36}$. Deduce expressions for the components of the dielectric polarization referred to the principal axes if a stress σ'_3 is applied normal to the (111) plate of the previous question. The cubic phase of ZnS (point group $\bar{4}3m$) has $d_{14} = 3.1 \times 10^{-12} \text{ C N}^{-1}$. Calculate the charges induced on a (111) plate of area 30 mm^2 . (*Note:* the obvious way to proceed is to transform the polarization tensor back into the reference axes of the plate, but a more direct method would be to transform the piezoelectric tensor $\mathbf{d}$ into the axes of the plate. It would be worth using both methods to give a check on the answers.)
- (7.6) The non-zero piezoelectric coefficients are quartz are $d_{11} = -d_{12}$, $d_{26} = -2d_{11}$, and $d_{14} = -d_{25}$. Data are given later in the book in Fig. 12.11. Calculate the strains produced when an electric field of strength 10^6 V m^{-1} is applied along the [100] direction, which corresponds to the x_1 axis, at room temperature.
- (7.7) Write the transformation matrix for a mirror plane, and use this to determine the non-zero components of the piezoelectric tensor d_{ijk} . Repeat the exercise for the case of a mirror plane at right angles to the original mirror

plane and hence determine the non-zero components for a crystal of point group $mm2$. Show that the addition of a third mirror plane causes all components of the piezoelectric tensor to be zero.

- (7.8) Write the transformation matrix analogous to eqn 7.53 for the operation of a $\bar{4}$ axis. Use the transformation matrix to obtain the non-zero values of the piezoelectric coefficients, and compare with those allowed in the presence of a 4-fold rotation axis.
- (7.9) Obtain the non-zero piezoelectric coefficients of a tetrahedron of point group symmetry $\bar{4}3m$, eqn 7.49, by consecutive application of the transformation matrices for the $\bar{4}$ along the three orthogonal axes.
- (7.10) The elastic energy is given by eqn 7.58. Show that for a cubic crystal, for which the only non-zero values of the elastic constant tensor are c_{11} , c_{12} , and c_{44} and their symmetrically equivalent components, stability is only achieved if $c_{44} > 0$, $c_{11} > |c_{12}|$ and $c_{11} + 2c_{12} > 0$. (*Hint:* for the second two conditions, one approach would be to write the equation for the energy as $E = \mathbf{e}^T \cdot \mathbf{c} \cdot \mathbf{e}$, where $\mathbf{e}$ is the column matrix with elements e_1, e_2, e_3 and $\mathbf{c}$ is a 3×3 matrix with the diagonal elements equal to c_{11} and the other elements equal to c_{12} . The condition for stability $E > 0$ is given by the condition that the eigenvalues of $\mathbf{c}$ are greater than 0.)
- (7.11) Show that the transformation of $\mathbf{P} \rightarrow \mathbf{P}'$ obtained from the applied electric field of eqn 7.11 is consistent with eqn 7.14.